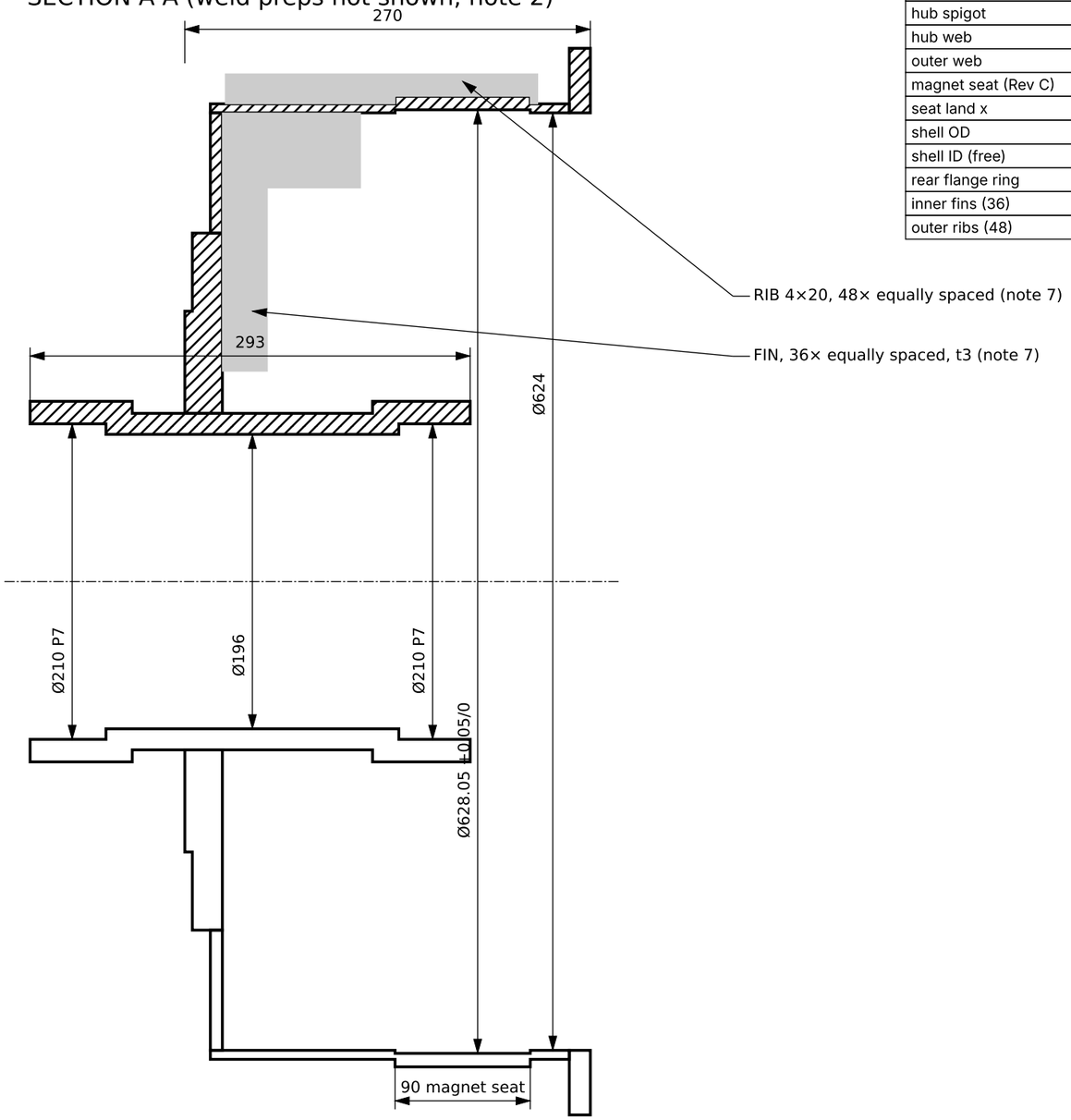
SECTION A-A (weld preps not shown, note 2)



Feature	Size	Tolerance / note
cup seats (2)	Ø210 × 45	P7, Ra 0.8, coax. 0.02
cup shoulders	Ø196	square 0.01 to axis
hub spigot	Ø360 × 5	f7; face flat 0.03
hub web	20 thick, Ø464	12× M16 thru PCD 400
outer web	8 thick	full pen. weld to shell
magnet seat (Rev C)	Ø628.05 × 90	+0.05/0, coax. 0.02, Ra 3.2
seat land x	335 to 425	as Rev B
shell OD	Ø646	steel under seat 8.98
shell ID (free)	Ø624	clearance to magnets
rear flange ring	Ø624/Ø710 × 14	M10 PCD 644 + M10 PCD 678
inner fins (36)	t3, x 220-312, r 140-312	L-plate, web + bore, every 10°
outer ribs (48)	4 × 20, x 222-430	tips Ø676, every 7.5°

NOTES

1. Rev C: magnet seat Ø628.05 +0.05/0 (was Ø618.40) for the 12 mm N45SH blocks; steel under the seat 8.98 (FEA: 1.45 T median, 1.7 T at the pole edges, included in the EMF).
2. Welds full penetration at web/housing and web/shell, ISO 5817 B, 100 % MT, UT of the shell seam (ring-rolled shell preferred, doc 18 §4.2). Stress relieve 580 to 620 °C before machining.
3. Machine in one set-up: cup seats Ø210 P7, magnet seat, hub spigot Ø360 f7, rear flange face; coaxiality ≤ 0.02, shoulders square ≤ 0.01.
4. Hub web 12× M16 thru PCD 400; rear flange ring 24× M10 PCD 644 (rear cover) and 24× M10 PCD 678 (disc).
5. Magnet seat: degrease, grit blast Sa 2½ just before bonding; no paint inside. Outside zinc spray + epoxy (C5).
6. Balance finished rotor (drum, magnets, sleeve, disc, cover) to ISO 21940-11 G2.5 at 250 rpm, correction on the rear flange ring.
7. Passive cooling (doc 23 §4a): 36 inner fins t3 S355, L-shaped, fillet-welded both sides to hub web and shell bore, every 10°, 9 mm clear of the end windings; 48 outer ribs 4 × 20 S355 on the shell outside, every 7.5°, intermittent fillet welds 30/60 both sides, ends rounded. Weld fins and ribs before stress relief and machining.
8. Rear flange ring welded to the shell full round, continuous: with the hub web it keeps the 9 mm shell round against the magnet pull (doc 23 §5). Seat roundness 0.05 after stress relief.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

ROTOR DRUM WELDMENT (GENERATOR YOKE + BEARING HOUSING)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-210C	C · 2026-11-05	1:3.8
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355J2+N, welded, stress relieved	91.8 kg + 12.5 kg fins/ribs	Claude / not checked